

ChargeEV - Final Project Document

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EV Charging Station Management Platform Complete project documentation · Everything in one place

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1. Project Overview

ChargeEV is a full-stack EV charging station management platform. It connects three types of users: **drivers** who need to charge their vehicles, **station owners** who list charging points, and **admins** who oversee the platform.

What makes it more than a simple booking app:

- **Real-time auctions** — when a charging slot is contested, drivers bid for it. A priority algorithm weighs bid amount against battery urgency.
- **Live updates** — Socket.IO pushes bid changes and notifications instantly, no page refresh.
- **Real payments** — Stripe integration with cost-locking at check-in and idempotent webhooks (a replayed webhook never double-charges).
- **KYC gating** — identity verification required before booking or listing.
- **Full audit trail** — every admin action is logged.

Project size

Metric	Value

Frontend source	~14,600 lines
Backend source	~6,900 lines
Files	45 <code>.jsx</code> , 24 <code>.js</code> , 2 <code>.css</code> (frontend) · 60 <code>.js</code> (backend)
Pages	26 (20 user-facing + 6 admin)
API endpoints	62 handlers across 13 route files
Database models	10 models, 6 enums
Shared components	16
Redux slices	13
Tests	73 passing, 5 suites
Git commits	16

2. Technology Stack

Frontend

- **React 18** with **Vite** as the build tool
- **Redux Toolkit** for state management (13 slices)
- **Bootstrap 5** for grid and layout
- **React Router** for navigation
- **Socket.IO client** for real-time updates
- **Stripe.js** for payments
- **Leaflet** for station maps
- **Recharts** for owner/admin analytics
- **Framer Motion** for page transitions
- **React Helmet Async** for SEO meta tags

Backend

- **Express** REST API
- **Prisma ORM** with **MongoDB**
- **Socket.IO** for real-time auction/bid delivery
- **Stripe** for payment processing
- **SendGrid** for transactional email
- **JWT** authentication (`jsonwebtoken` + `bcryptjs`)
- **Helmet**, **CORS**, **express-rate-limit** for security
- **express-validator** for input validation
- **Sentry** for error tracking (optional, dormant until configured)

Tooling

- **ESLint** + **Prettier** (both configured from scratch during this engagement)
- **Jest** + **Supertest** for testing

- Google Analytics 4 (optional, dormant until configured)

3. Architecture & Folder Structure

```
ev-management/
├── frontend/                React 18 SPA (Vite)
│   ├── src/
│   │   ├── pages/          26 route-level pages (incl. pages/admin/)
│   │   ├── components/     16 shared components
│   │   ├── store/          Redux Toolkit – 13 slices + cache layer
│   │   ├── utils/          9 helpers (api, socket, logger, analytics...)
│   │   └── styles/         grid.css (staged for Phase 10)
│   ├── backend/           Express REST API + Socket.IO
│   │   ├── controllers/    11 controllers (business logic)
│   │   ├── routes/         13 route files (62 handlers)
│   │   ├── middleware/     auth, error, kyc, validate, validateQuery
│   │   ├── utils/          15 helpers (prisma, jwt, email, stripe...)
│   │   ├── prisma/         schema.prisma + seed.js
│   │   ├── tests/          5 suites, 73 tests (in-memory mock DB)
│   │   └── scripts/        doctor.mjs (preflight health check)
│   ├── docs/
│   │   ├── screenshots/    Design mockups
│   │   └── pdf/            PDF versions of the reports
│   └── scripts/            setup-env.mjs (generates .env with secrets)
```

Request flow

```
React page
├─> Redux thunk
│   └─> utils/api.js (axios)
│       └─> Express route
│           └─> validation middleware
│               └─> auth / KYC middleware
│                   └─> controller
│                       └─> Prisma
│                           └─> MongoDB
```

Real-time updates (auctions, bids, notifications) bypass this entirely and push over **Socket.IO**.

Shared components

AddressAutocomplete, AdminLayout, BackToTop, ErrorBoundary, Footer, JsonLd, Marquee, Navbar, OfflineBanner, PasswordStrengthBar, PaymentModal, SEO, Skeleton, Spinner, Stars, StationsMap

4. Data Model

Models (10)

Model	Purpose

User	Drivers, owners, and admins (differentiated by Role)
EV	A driver's registered electric vehicles
ChargingStation	Owner-listed charging locations (needs admin approval)
Slot	Individual charging points within a station
Booking	A time-window reservation of a slot
Bid	An auction bid on a contested slot
Payment	Stripe payment records
Review	Station reviews (verified purchases only)
Complaint	User-submitted issues → admin inbox
Log	Audit trail of admin actions

Enums (6)

Role · StationStatus · SlotStatus · BookingStatus · BidStatus · PaymentStatus

Core relationships

A **User** (driver) owns **EVs** and makes **Bookings** against a **Slot** belonging to a **ChargingStation** owned by another **User** (owner). Contested slots go to auction via **Bids**. Completed bookings unlock **Reviews**. **Payments** tie to Stripe. **Logs** power the admin audit trail.

5. User Roles & Permissions

Role	Capabilities
Driver	Register EVs · browse & book stations · bid in auctions · pay via Stripe · leave reviews · file complaints
Owner	Everything a driver can do, plus: create stations & slots (pending admin approval) · view revenue reports · close auctions
Admin	Approve/reject stations · manage users · resolve complaints · view dashboard & full audit log

Enforcement: - `auth.middleware.js` — JWT verification and role guards - `kyc.middleware.js` — gates booking and station listing behind identity verification

6. API Reference (62 endpoints)

Route file	Count	Covers
admin	9	Dashboard, user/station management, audit logs
auth	8	Register, login, logout, password reset, admin setup
booking	7	Create, check-in, complete, cancel, history
slot	7	Slot CRUD, availability
station	7	Station CRUD, public listing, approval
bid	5	Place bid, auction close, priority scoring
ev	5	User's EV CRUD
verification	4	Email OTP, KYC
complaint	3	Submit, admin inbox, resolve
review	3	Verified-purchase reviews
payment	2	Stripe payment intent + webhook
ai	1	Station recommendation
user	1	Profile

Full request/response detail for every endpoint is in [API.md](#).

7. Key Features

Booking system

Time-window reservations with **overlap detection** and **race-condition safety** — verified under real parallel load, not just sequential calls. Check-in locks the cost so pricing can't shift mid-session.

Auction system

When a slot is contested, drivers bid. Priority score is:

$$60\% \times (\text{normalized bid amount}) + 40\% \times (\text{battery urgency})$$

Bids are normalized against a **fixed reference ceiling**, not the running maximum bid. This matters: normalizing against “current max” made scores order-dependent — the first bidder always scored 1.0 regardless of bid size, because their own bid was both numerator and denominator. A fixed ceiling keeps rankings stable no matter what order people bid in.

Auctions close automatically when their deadline passes (previously they could stay stuck open forever if the owner never manually closed them).

Payments

Stripe integration with **idempotent webhooks** — a replayed webhook charges exactly once, verified by test.

Real-time

Socket.IO pushes bid updates, auction closures, and notifications live.

Security hardening (from Phase 7)

- **Stored XSS fixed** — user names were interpolated raw into email HTML with zero escaping. Now auto-escaped via a tagged template across all 12 email templates.
- **Data leak fixed** — the public station list was exposing `totalRevenue` and `ownerId` to every visitor. Fixed with explicit field selection.
- **Magic-byte image validation** — uploads are checked by actual file signature, not just extension.

Accessibility (from Phase 7.3)

- Real computed WCAG contrast math (not eyeballed) — genuine failures found and fixed
- Shared `Modal` rebuilt with proper dialog ARIA, Escape-to-close, and focus management (used by all 13 modals app-wide — none of this existed before)
- 58 form labels across 11 files fixed with proper `htmlFor` association

SEO (from Phase 4)

Meta tags on all 26 pages, JSON-LD structured data (Organization, Product, Review schemas), `sitemap.xml`, `robots.txt`, corrected heading hierarchy.

8. Engineering Decisions Worth Knowing

These came up repeatedly. Read them before changing related code.

Caching without new infrastructure

The tech-stack constraint ruled out React Query and Redis. So instead: - **Frontend**: a ~30-line TTL cache using Redux Toolkit's own `createAsyncThunk` `condition` `option` (`store/cacheCondition.js`) - **Backend**: a `Map`-based in-memory TTL cache (`utils/simpleCache.js`)

Both are **deliberately scoped to low-risk targets only** (station list, My EVs, public station list). They are *not* applied to pages with live Socket.IO listeners or admin correctness requirements — time-based caching is a bad fit there.

Bundle analysis

`webpack-bundle-analyzer` cannot attach to this project — it uses Vite/Rollup, not Webpack. Used `rollup-plugin-visualizer` instead, gated behind an opt-in `npm run build:analyze` script so the analysis artifact never ships in a production build.

Logging discipline

All logging routes through `logger.js` on both sides. `logger.debug` is dev-only. Example: an expired JWT in

`auth.middleware.js` uses `logger.debug`, not `logger.error` — it's a routine, expected event, not a server problem, so it shouldn't create production log noise.

Graceful boot failures

The server runs a `SELECT 1` against the database **before** it starts listening. An unreachable DB now fails at boot with an actionable checklist instead of surfacing as an opaque 500 on the first request. `npm run doctor` diagnoses env vars, Prisma client generation, and DB connectivity — each failure paired with its specific fix.

Sentry integration point

Sentry is wired into `logger.js`'s `error` method on both frontend and backend. Because `error.middleware.js` already funnels every unhandled route error through `logger.error`, that single hook gives the entire app remote error visibility for free.

9. Work Completed (Phases 1-10)

Phase	Outcome
1	Structure audit · dev-gated logger · PropTypes on 18 components · env setup script · fixed broken dev proxy
2	Full rebrand — cream/gold/dark palette (<code>#FDF8F0</code> / <code>#C9A96E</code> / <code>#1A1A1A</code>) · new hero · CSS variables (legacy names kept as aliases, ~40 files depend on them)
3	CTA audit · 44px minimum touch targets · CTAs added to empty states · micro-copy
4	SEO — meta tags on all 26 pages · JSON-LD · sitemap · robots.txt · heading hierarchy
5	Feature verification · dead-code report · fixed systemic silent-failure bug (4 places where a failed fetch looked identical to “no data”) · built AddressAutocomplete, password-strength indicator, offline detection · mobile audit (found iOS Safari auto-zoom bug: inputs were 15.2px, fixed to 16px)
6	Performance — WebP images · self-hosted fonts (enabling real preload) · surgical <code>React.memo/useMemo</code> · DB indexes · fixed public endpoint leaking revenue data · in-memory caching
7	Testing/QA — genuine concurrent-request race test · stored-XSS fix · magic-byte image

	validation · WCAG contrast math · Modal ARIA rebuild · 58 form-label fixes
8.1	Documentation — README rewrite · <code>API.md</code> (65 endpoints) · deployment guide
8.2	Code quality — ESLint + Prettier configured from scratch · console.log audit · naming conventions · fixed a real socket-subscription bug in <code>AuctionHub.jsx</code>
8.3	Deployment prep — real DB-backed health check · Sentry · Google Analytics 4
9	Final verification — fixed 3 crashing test suites · full 13-item checklist audit
10	Bootstrap removal — scoped and staged, deliberately paused (see §13)

Full chronological detail: `CHANGELOG.md` (most recent first).

10. Current Status — Tests, Bugs & Gaps

All automated checks pass

Check	Result
Backend test suite	73/73 passing, 5/5 suites
Backend ESLint	0 problems
Frontend ESLint	0 problems
Frontend build	succeeds
Import resolution	313/313 resolve, case-exact
Backend syntax (<code>node --check</code>)	all files parse
<code>npm audit</code> (backend)	0 vulnerabilities
TODO/FIXME markers in source	none

No known bugs, broken imports, syntax errors, or failing tests.

Functionality: verified vs. not

Verified working (backed by passing tests)

- Admin bootstrap · registration & login for every role · role guards
- Station lifecycle (owner creates → admin approves)
- Time-window bookings **with overlap/race-condition rejection**
- Check-in cost-locking · Stripe webhook idempotency · completion
- Full auction flow (bidding, priority scoring, closing, auto-booking)
- Admin dashboard & audit trail · complaints · verified-purchase reviews
- Logout revoking tokens server-side

Present but not directly tested

Not broken — just not covered by an automated assertion: - AI recommendation endpoint - Email OTP happy path (validation errors *are* tested) - Live Socket.IO delivery to a connected client - EV and slot CRUD as standalone endpoints

Cannot be verified in this environment

No live browser was available at any point in this engagement: - Browser console errors - Real click-through 404 testing - Mobile rendering at actual viewport widths - Visual design polish

Known open items (deliberate, not bugs)

1. **Bootstrap removal paused mid-plan** — `grid.css` is written but *not imported*; Bootstrap is still fully active. The app is in a consistent state; the migration simply hasn't touched any page yet.
2. **DRY / component-extraction pass** — last unfinished Phase 8.2 item.
3. **Sentry & GA are dormant** — they no-op until you supply `SENTRY_DSN` / `VITE_GA_MEASUREMENT_ID`.
4. **Optional Dockerfile** — not created (Vercel/Render don't need it).
5. **~11 hardcoded hex colors** in components (star ratings, status badges) sit outside the shared CSS-variable palette — a consistency smell, not a bug.
6. **Frontend esbuild / vite advisory** — dev-server only, requires a breaking Vite major bump, intentionally deferred.

11. How to Run the Project

Step 1 — Generate environment files

```
node scripts/setup-env.mjs
```

This creates both `.env` files with real random secrets (JWT signing key, admin setup key). Safer and faster than copying `.env.example` by hand.

Step 2 — Fill in your credentials

In `backend/.env` :

Variable	Required?	Notes
<code>DATABASE_URL</code>	Yes	Your MongoDB / Atlas connection string
<code>SENDGRID_API_KEY</code>	Optional	Leave empty to disable email

STRIPE_* keys	Optional	Only if PAYMENT_MODE="live"
SENTRY_DSN	Optional	Leave empty to disable error tracking

In `frontend/.env` (both optional): `VITE_SENTRY_DSN` · `VITE_GA_MEASUREMENT_ID`

Step 3 — Backend

```
cd backend
npm install      # also runs `prisma generate`
npm run doctor   # preflight: env vars, Prisma client, DB connectivity
npm start        # → http://localhost:5000
```

`npm run doctor` is worth running first on any new machine — it checks everything needed to boot and tells you exactly what to fix if something's missing.

Step 4 — Frontend

```
cd frontend
npm install
npm start      # → http://localhost:3000
```

Admin panel lives at `/admin/*`.

Verify anytime

```
cd backend && npm test      # expect: 73/73 passing
cd frontend && npm run build # should complete cleanly
```

The test suite runs against an **in-memory mock database** — no MongoDB or `.env` required. It's the primary regression check.

12. Deployment Guide

Target architecture: Vercel (frontend) + Render (backend) + MongoDB Atlas (database).

Both Vercel and Render provide HTTPS automatically for their URLs — nothing in the app's own code assumes plain HTTP.

The backend exposes a `/health` endpoint that runs a real database query and returns **503** on failure (not just a static 200), already wired into Render's health-check configuration.

Complete step-by-step instructions — including Atlas setup, environment variables for each platform, first-admin creation, and troubleshooting — are in `DEPLOYMENT.md`.

13. Remaining Work & Recommendations

Priority 1 — Verify what this environment couldn't

Boot the backend on a real machine with a real database. Run `npm run doctor`, then `npm start`. This sandbox had no internet access to Prisma’s servers, so the query-engine binary could never download and the backend could never fully boot here. The code is standard and should work fine — but this is the single most important thing to confirm before considering deployment verified.

Open the app in a real browser with DevTools. This clears the entire “cannot verify” category at once — console errors, real 404 navigation, mobile rendering, and visual polish.

Priority 2 — Finish the open items

- **DRY pass / component extraction** — the last Phase 8.2 checklist item
- **Bootstrap removal** — resume via the pilot-page approach in `PHASE_10_BOOTSTRAP_REMOVAL_PLAN.md`. Important: migrate one low-traffic page first and check it visually before touching the other 33 files. This is the one category of change that no test, linter, or build can verify — only your eyes can.

Priority 3 — Nice to have

- Add test coverage for the untested endpoints (AI recommend, OTP happy path, live sockets)
- Centralize the ~11 hardcoded hex colors into CSS variables
- Consider the Vite major-version bump to clear the `esbuild` advisory (on its own schedule — it’s a breaking change)
- Replace the design mockups in `docs/screenshots/` with real screenshots once the app is running in a browser

14. Document Index

All documents live at the repository root. PDF versions of the main reports are in `docs/pdf/`.

Document	Use it for
<code>FINAL_PROJECT_DOCUMENT.md</code>	This document — everything in one place
<code>CURRENT_STATE_REPORT.md</code>	Bugs, gaps, functionality status
<code>CODEBASE_SUMMARY.md</code>	Condensed technical orientation
<code>README.md</code>	Setup, tech stack, design, security
<code>DEPLOYMENT.md</code>	Vercel + Render + Atlas, step by step
<code>API.md</code>	Every endpoint, exhaustively
<code>CHANGELOG.md</code>	Full engagement history, most recent first
<code>PHASE_10_BOOTSTRAP_REMOVAL_PLAN.md</code>	Resuming the CSS migration
<code>PHASE_9.2_FINAL_REPORT.md</code>	Changes, issues fixed, recommendations
<code>PHASE_7.1_TEST_PLAN.md</code>	What’s live-tested vs. code-traced
<code>PHASE_7.2_EDGE_CASES_REPORT.md</code>	Race conditions, XSS fix, uploads
<code>PHASE_7.3_ACCESSIBILITY_REPORT.md</code>	Contrast math, Modal rebuild

PHASE_6.1 / 6.2 / 6.3 reports	Performance & build decisions
PHASE_5.x reports	Feature verification, dead code, mobile audit
PROJECT_SUMMARY.md , SETUP_GUIDE.md	Original pre-engagement docs
CHANGELOG_FIXES.md	Prior engagement's history (untouched)

End of document.